

Establishing the limit of detection of laser-induced breakdown spectroscopy for *Escherichia coli* in artificial cerebrospinal fluid for the diagnosis of bacterial meningitis

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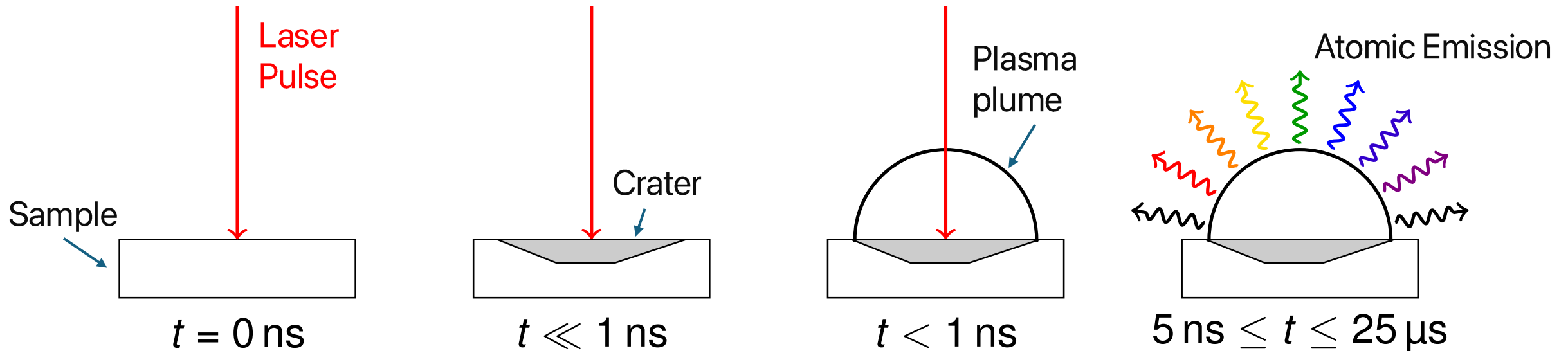
Bacterial Meningitis

- Life-threatening inflammatory infection of the meninges
- Lumbar puncture → CSF sample → protein/glucose/**bacteria content** → diagnosis
- Odds for unfavourable outcomes can increase by up to 30%/hour¹
- Need **single, quasi-instantaneous**, and **accurate** test for bacterial presence in CSF

¹Køster-Rasmussen, R., Korshin, A., and Meyer, C. N., "Antibiotic treatment delay and outcome in acute bacterial meningitis," *Journal of Infection*. **57**, 449 (2008).

Laser-Induced Breakdown Spectroscopy

- Sample ablated into plasma ($T = 10^4 \text{ K}$, $n_e = 10^{17} \text{ e}^-/\text{cm}^3$)
- Atomic emission gives elemental composition of sample
- Ability to assay **multiple** elements, **quasi-instantaneously** on samples with **minimal sample preparation**



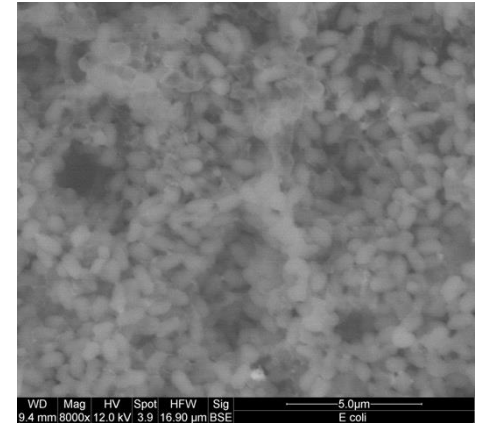
Classification accuracy of **suitably dilute bacterial pathogens** in CSF needs to be ascertained.

In bacterial meningitis cases, concentrations can be 10^3 – 10^8 CFU/mL.²

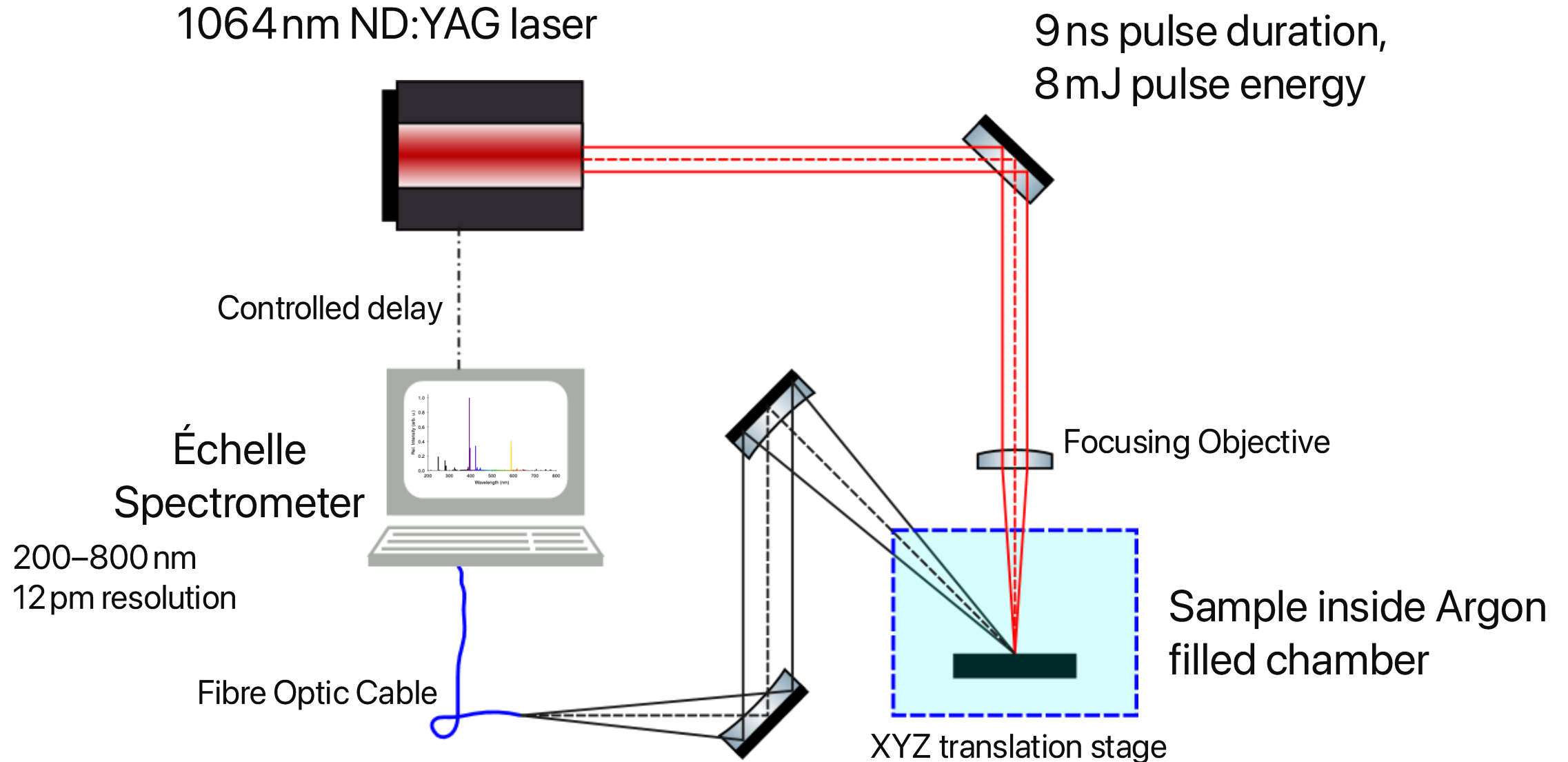
²Feldman, W. E., "Concentrations of bacteria in cerebrospinal fluid of patients with bacterial meningitis," *The Journal of Pediatrics*. **88**, 549 (1976).

Experimental Setup

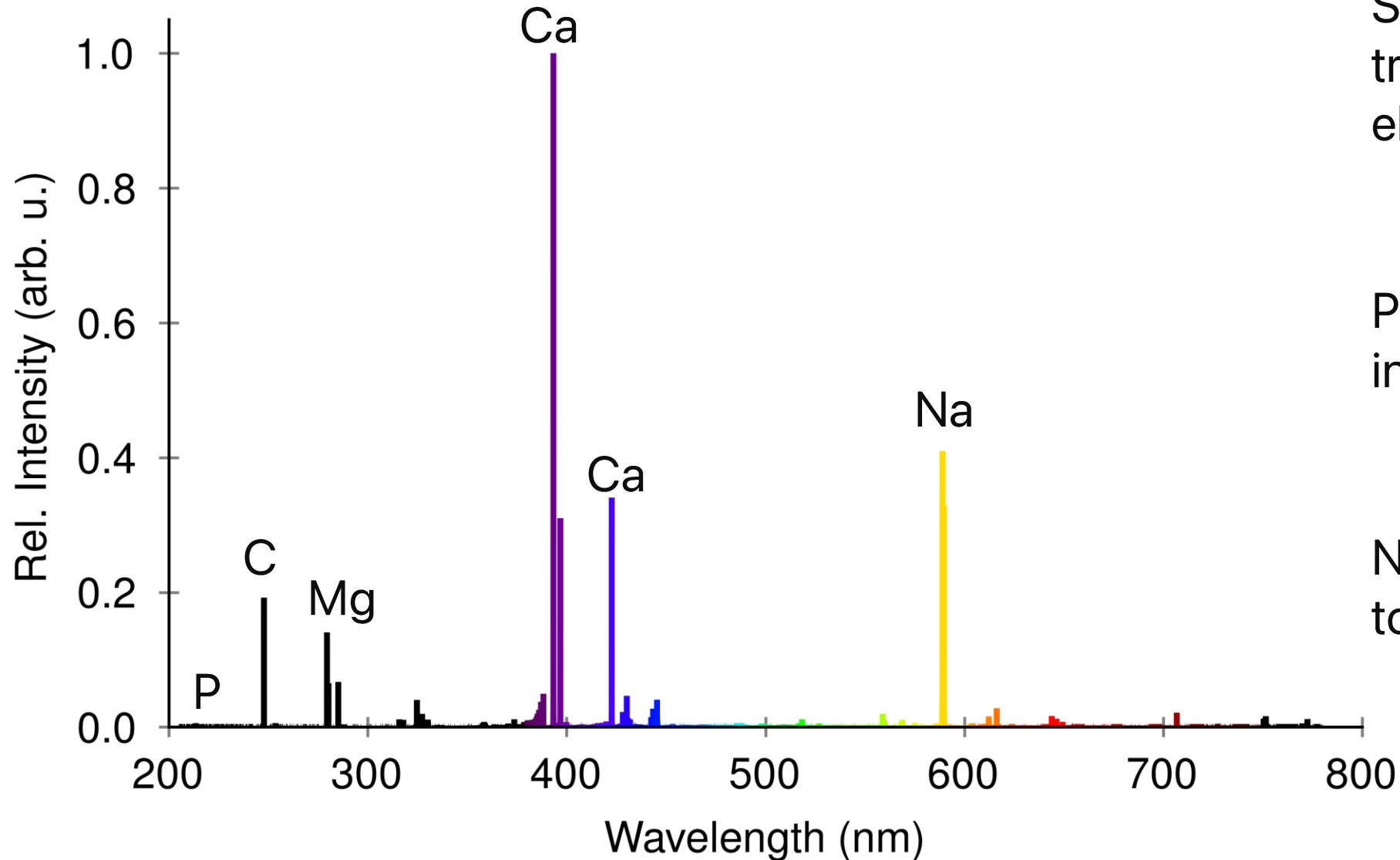
- Samples of *E. coli* made in artificial cerebrospinal fluid (aCSF)
- Optical densitometry to confirm concentrations
- 100 μ L of *E. coli* solution concentrated on 0.45 μ m nitrocellulose "filter"
- Concentrations of 26500, 5300, 1330, 1060, 530, 270, and 180 CFU/laser shot



Experimental Setup



LIBS Spectra

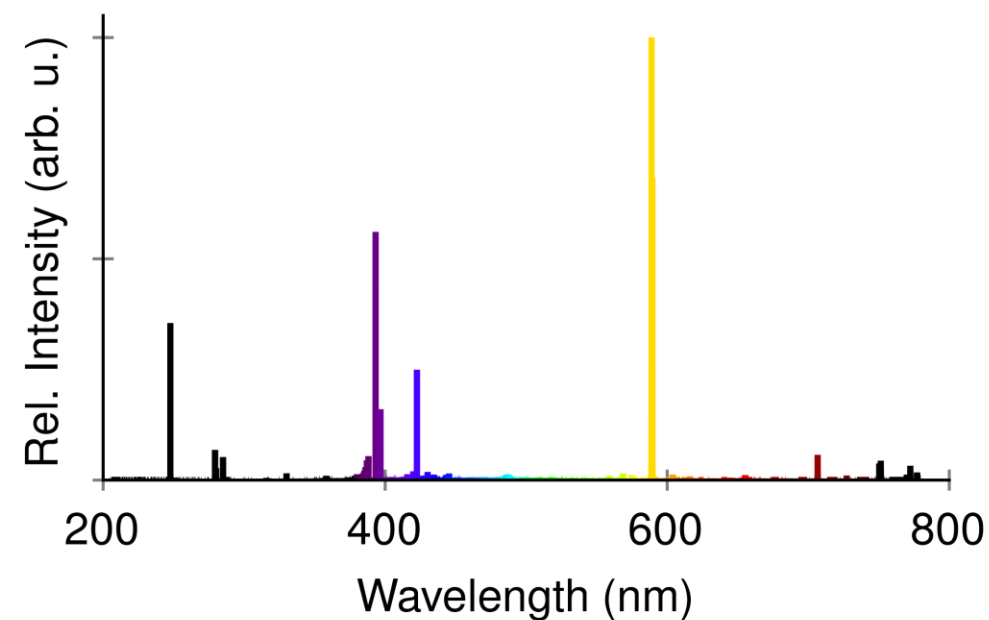
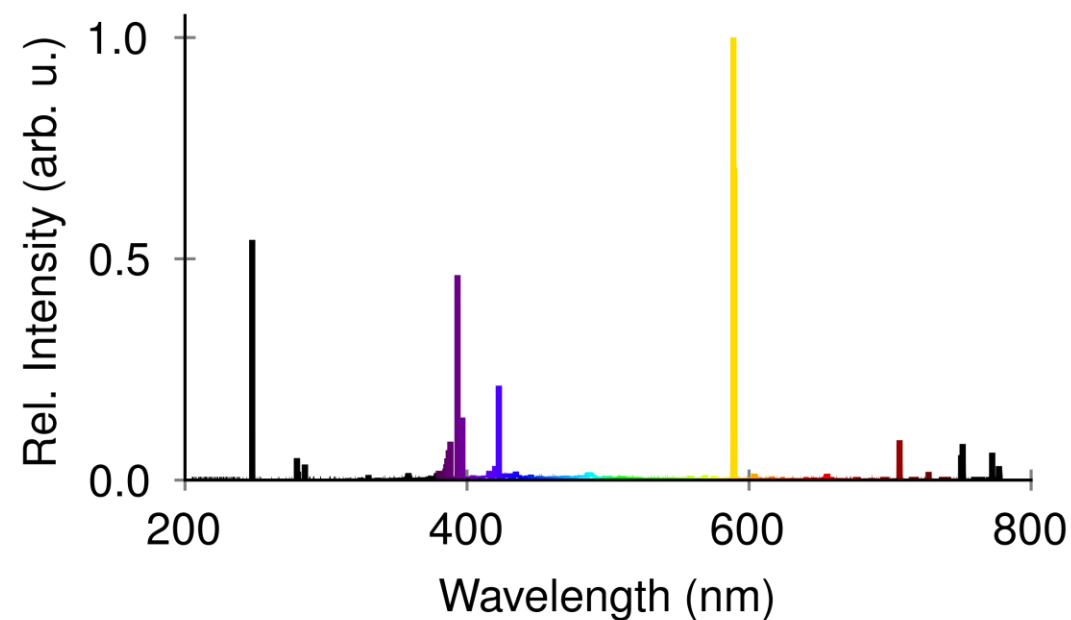
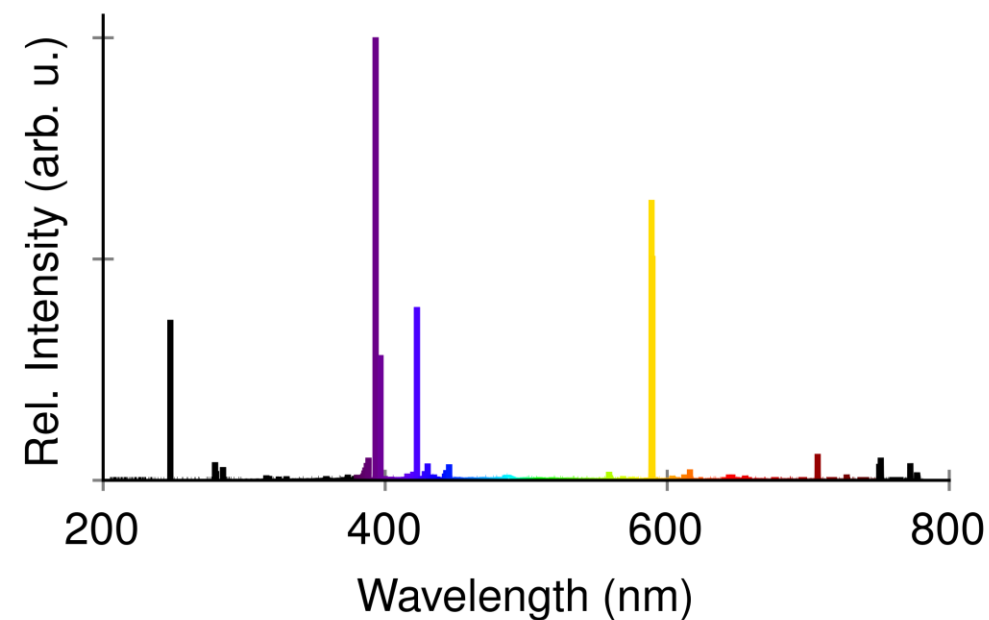
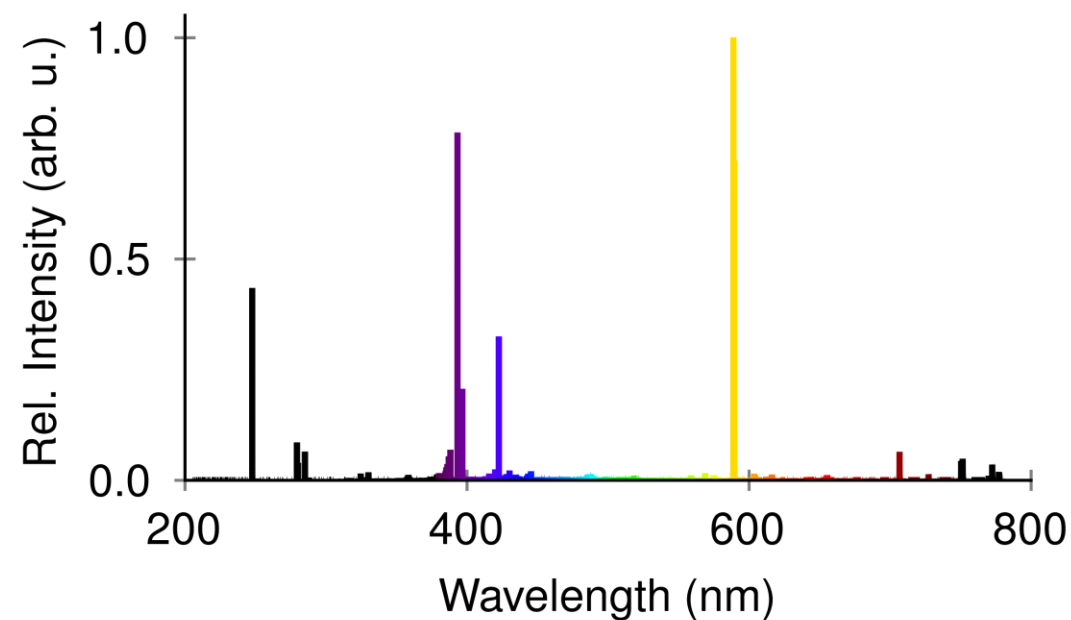


Signal comes from
trace inorganic
elements in the cells

P and Zn lines often
indicative of bacteria

Need systematic way
to make diagnoses

LIBS Spectra



Partial Least Squares-Discriminant Analysis

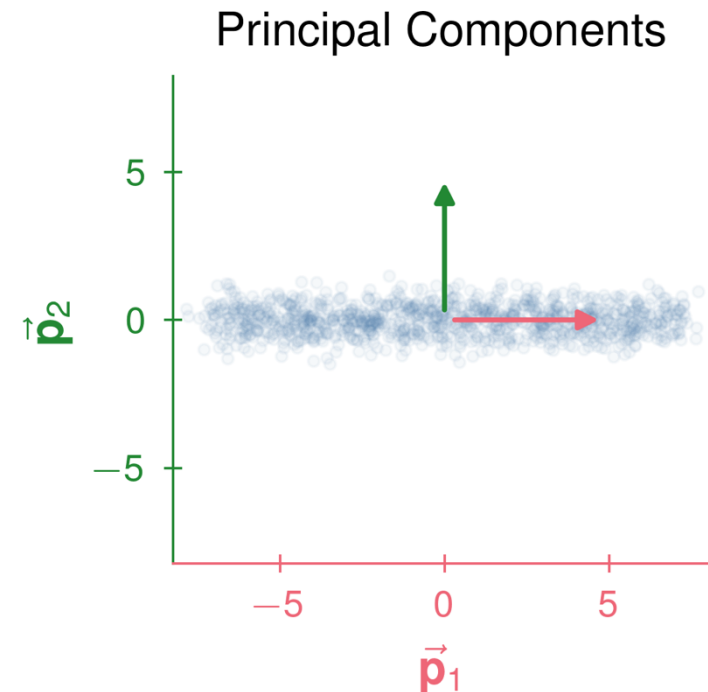
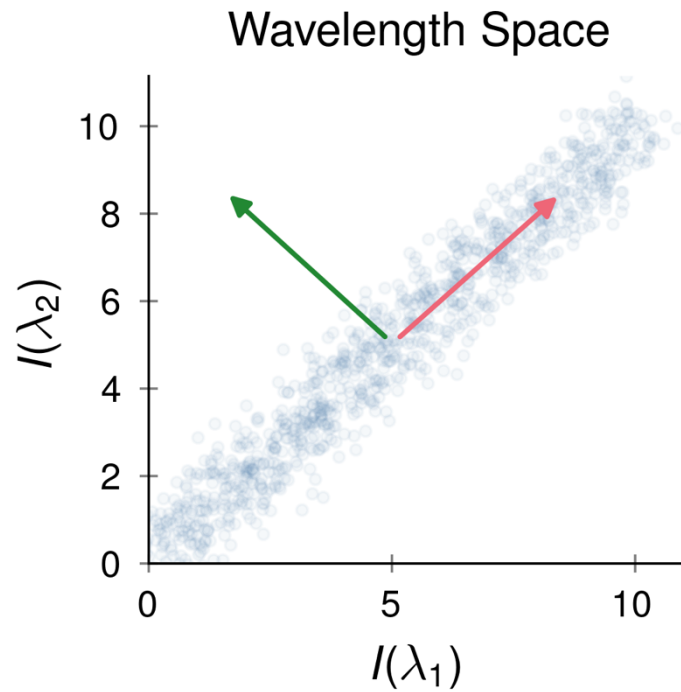
- Test on 15 integrated emission line intensities and 92 ratios of them
- All spectra from each filter were averaged together to increase SNR
- Latent variables constructed to maximize covariance of bacterial presence and intensity
- **Limit of detection: ~5300 CFU/shot**

Concentration (CFU/shot)	Sensitivity
26500	80%
5300	100%
2650	50%
1330	43%
1060	0%
530	0%
270	10%
180	0%

Principal Component Analysis

- 10 principal components instead of 50 000 wavelength variables
- Captures 97% of the variance

Simulation:



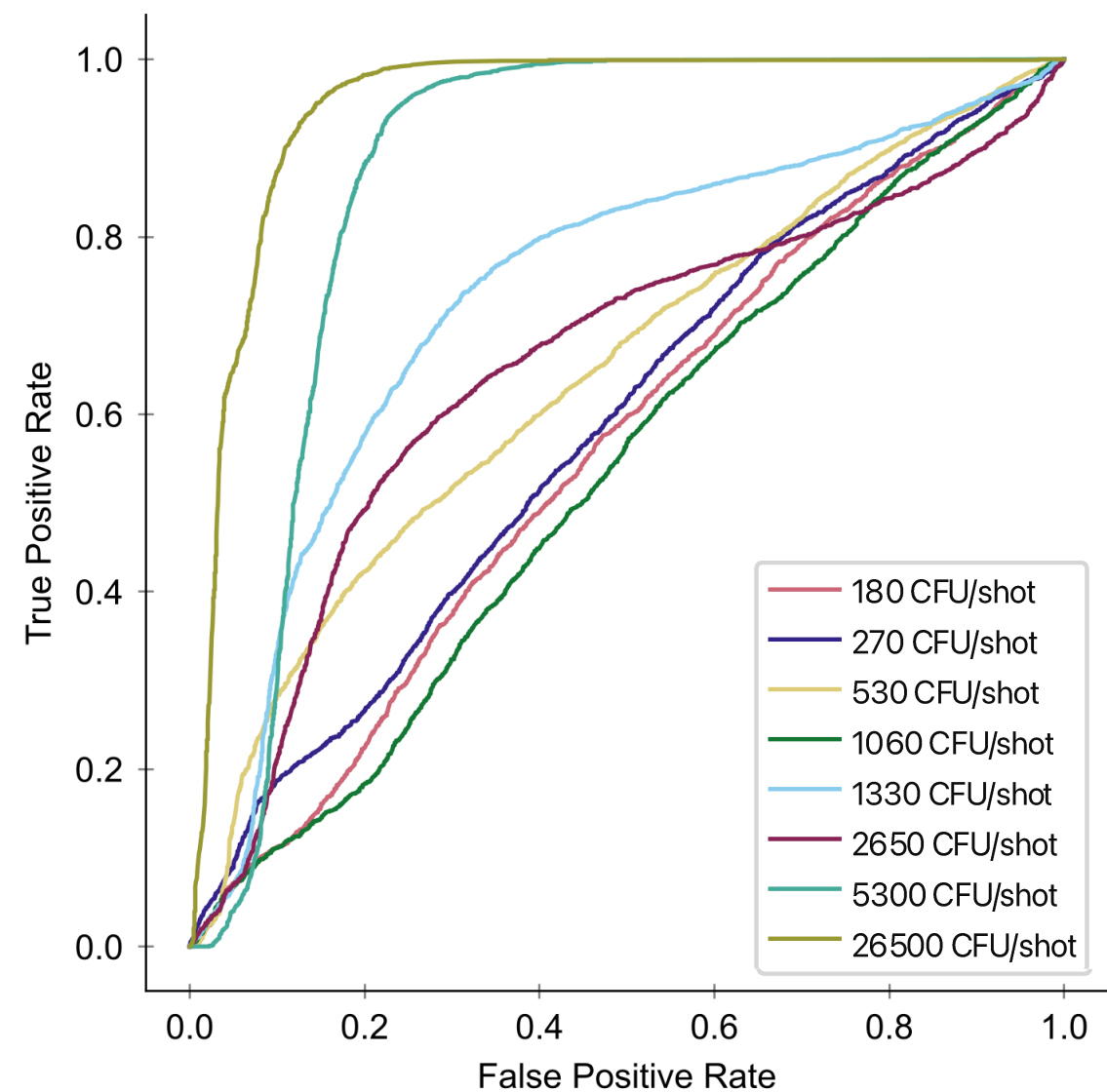
$\vec{p}_1$ explains most of the variance
Can disregard $\hat{p}_2$

Artificial Neural Network

- ANN made with TensorFlow's *Keras* API
- 10 PCA components as input layers
- 1 hidden layer
- Most probabilistic threshold
- **Limit of detection: ~1330 CFU/shot**

Concentration (CFU/shot)	Classification Accuracy
26500	99%
5300	100%
2650	77%
1330	91%
1060	61%
530	57%
270	60%
180	68%

Artificial Neural Network



The **ROC** (receiver operating characteristic) **curve** is an alternate, traditional measure of a test's diagnostic ability

Concentration (CFU/shot)	Area Under the ROC Curve
26500	95%
5300	87%
2650	65%
1330	73%
1060	54%
530	64%
270	58%
180	56%

Comparing Results

Concentration (CFU/shot)	PLS-DA Sensitivity	PCA-ANN Classification Accuracy	PCA-ANN Area Under the ROC Curve
26500	80%	99%	95%
5300	100%	100%	87%
2650	50%	77%	65%
1330	43%	91%	73%
1060	0%	61%	54%
530	0%	57%	64%
270	10%	60%	58%
180	0%	68%	56%

Conclusions & Future Work

Conclusions

- I have simulated bacterial meningitis infections using *E. coli* in aCSF
- I have determined the classification accuracy of LIBS on varying concentrations
- **LIBS can detect bacteria when 10^3 CFU/laser shot are present**

Future Work

- Test different species and make more reproducible concentrations
- Test on patient-acquired CSF
- Investigate effects of UV (or OPO) laser pulse to enhance emission

Thank You!



Bonus Slides

Learn More

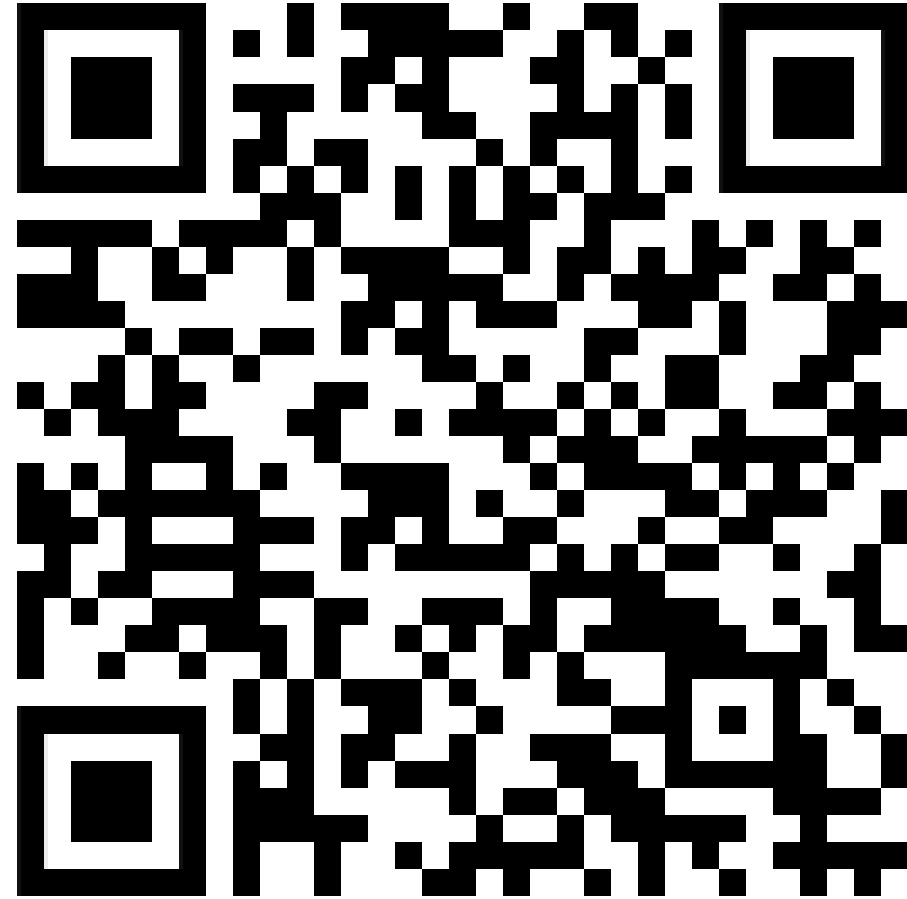
E.J. Blanchette. *Detection and Diagnosis of Bacterial Pathogens in Blood and Urine Using Laser-Induced Breakdown Spectroscopy*. Master's thesis, University of Windsor, 2022.

E.J. Blanchette et al., "Detection and Classification of Bacterial Cells After Centrifugation and Filtration of Liquid Specimens Using Laser-Induced Breakdown Spectroscopy," *Applied Spectroscopy* **76**, 2022, pp. 894-904. <https://doi.org/10.1177/000370282210927>

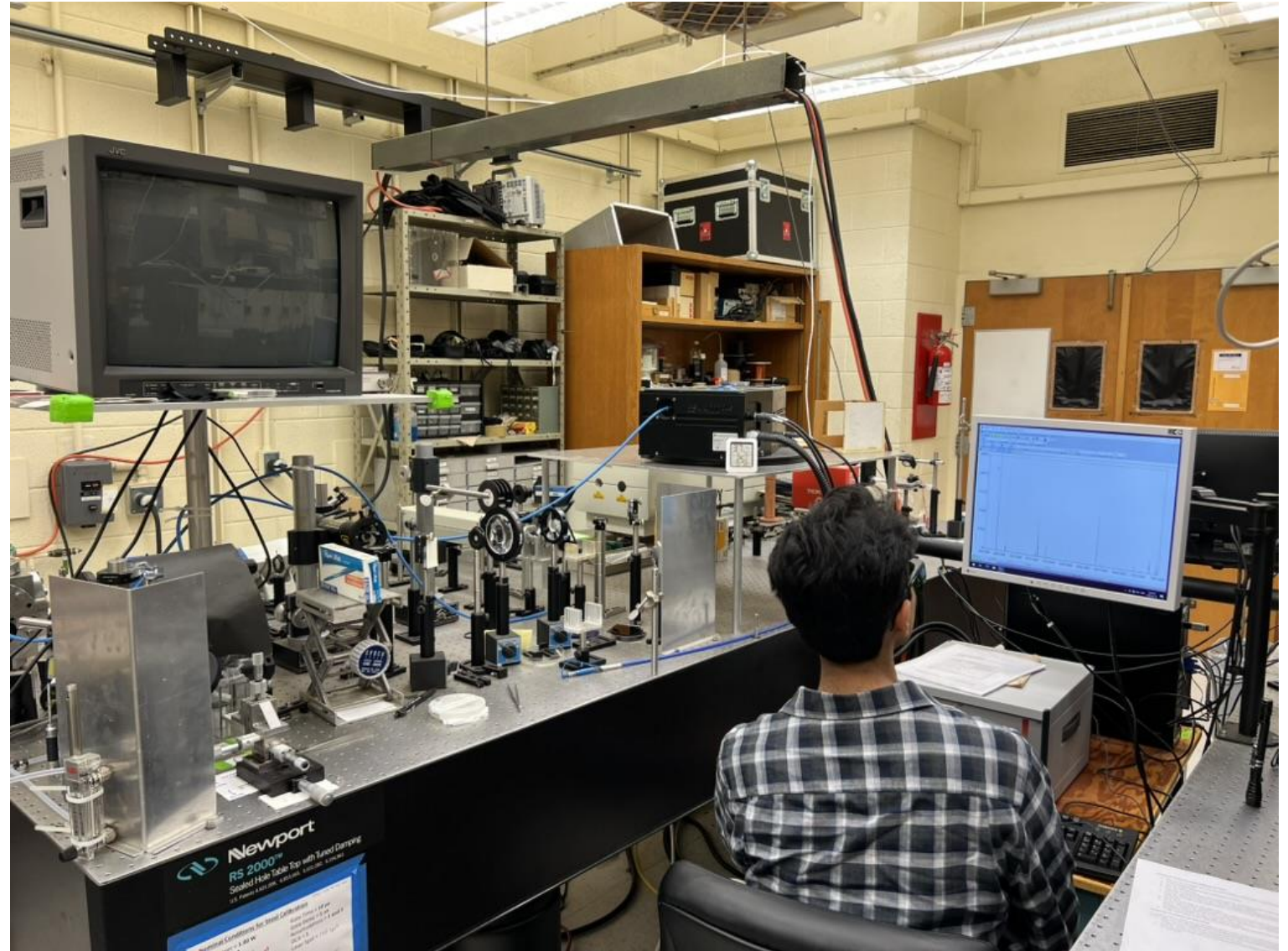
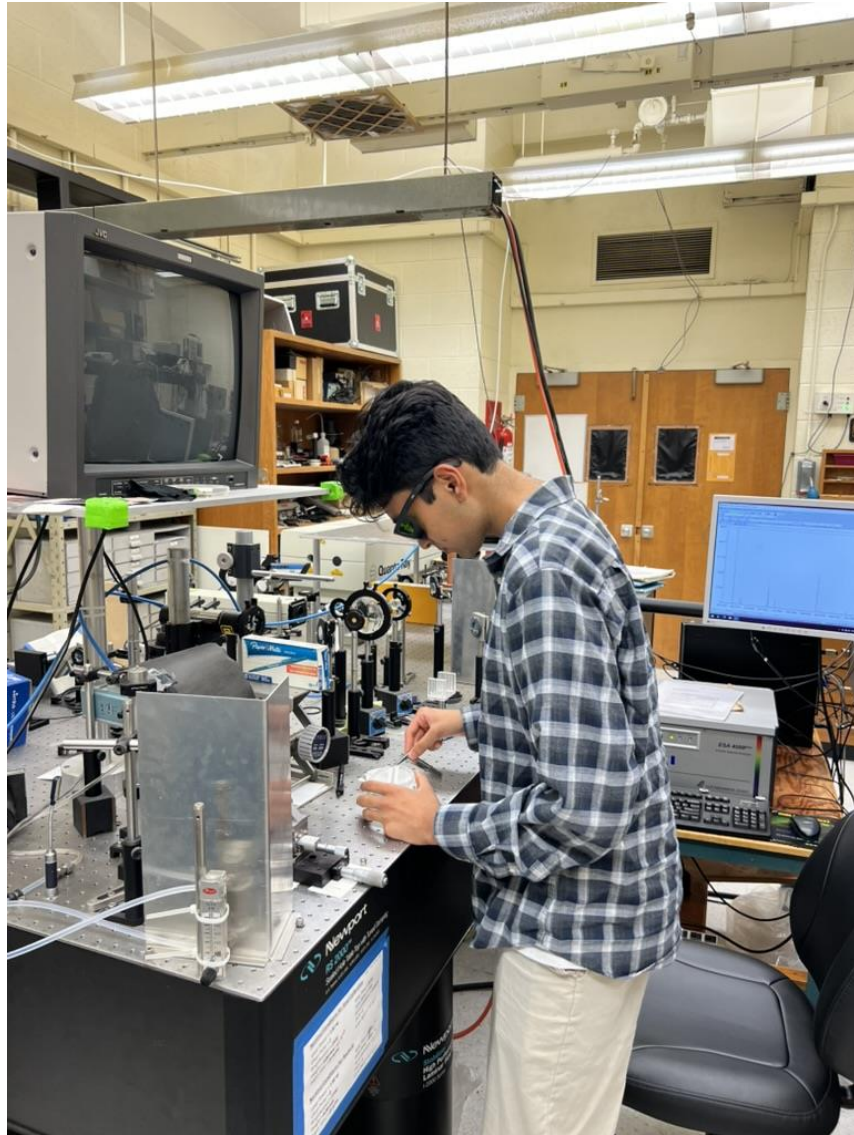
E.J. Blanchette, E.A. Tracey, A. Baughan, G.E. Johnson, H. Malik, C.N. Alionte et al. "Detection and diagnosis of bacterial pathogens in blood using laser-induced breakdown spectroscopy," *Spectrochim. Acta B* **215** (2024) 106911. <https://doi.org/10.1016/j.sab.2024.106911>

E.J. Blanchette, E.A. Tracey, A. Baughan, G.E. Johnson, H. Malik, C.N. Alionte, et al. "Detection and diagnosis of bacterial pathogens in urine using laser-induced breakdown spectroscopy," *Spectrochim. Acta B* **216** (2024) 106944. <https://doi.org/10.1016/j.sab.2024.106944>

Access full list of publications:

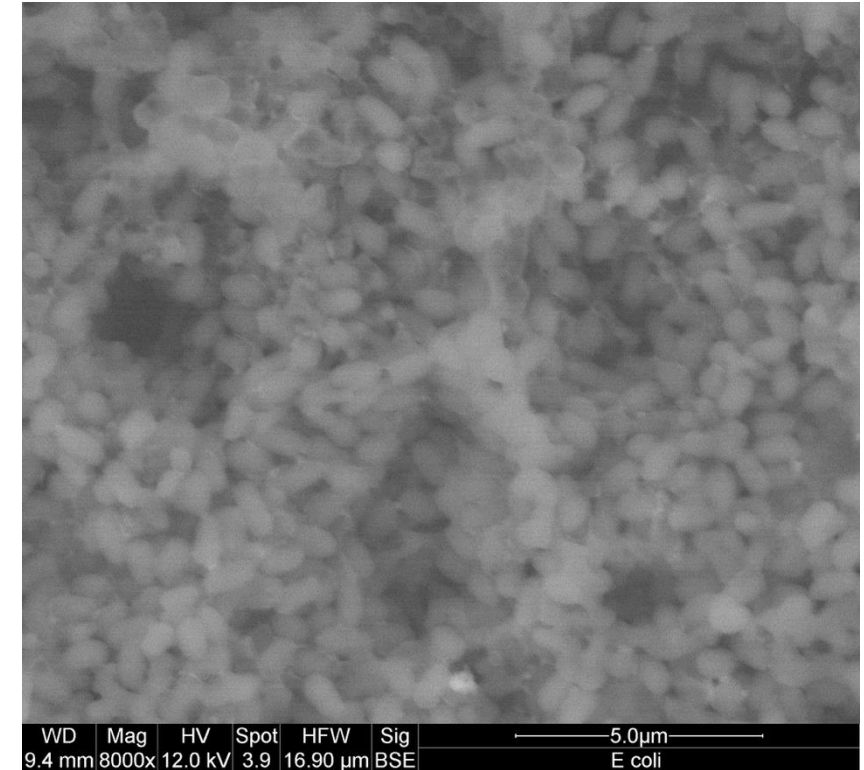


Our Lab



The *E. coli* bacterium

- 1–2 μm in length, 0.25 μm in diameter
- Most strains do not contain disease; many are found in the gut
- Some strains are responsible for **neonatal meningitis**, UTIs, Crohn's disease



SEM image of *E. coli* colony

Cerebrospinal Fluid

- CSF is a clear, transcellular, nominally sterile fluid that surrounds the brain, spinal cord; found within the meninges.
- Bacteria within CSF is often indicative of bacterial meningitis.
- Artificial CSF (aCSF) is a fluid designed to mimic the ionic concentrations of CSF, and is the fluid used in this study.

Ion	Concentration (mM) in aCSF
Na ⁺	150
K ⁺	3.0
Ca ²⁺	1.4
Mg ²⁺	0.8
pH ³⁻	1.0
Cl ⁻	155

PCA versus PLS-DA

PCA	PLS-DA
Finds direction of maximum variance in the spectral intensities X	Finds directions of maximum covariance between spectra X and bacterial presence Y
Maximizes $\text{var}(\mathbf{X})$	Maximizes $\text{cov}(\mathbf{X}, \mathbf{Y})$
(Eigen)decomposition of X	Projects both X and Y to new spaces
"Unsupervised" algorithm	"Supervised" algorithm
Preprocessing before discriminatory ANN	Both preprocessing and discrimination

The Plasma in Time

- LIBS is temporally-resolved in that the spectrum of the plasma changes through its lifetime.
- At early times:
 - Degree of ionization (ratio of electrons to other species) is high
 - Presence of background continuum from bremsstrahlung and recombination events
- At later times:
 - Background continuum diminishes faster than spectral lines
 - Line shapes narrow

LIBS in Liquids

LIBS **can** be performed on any state of matter, but

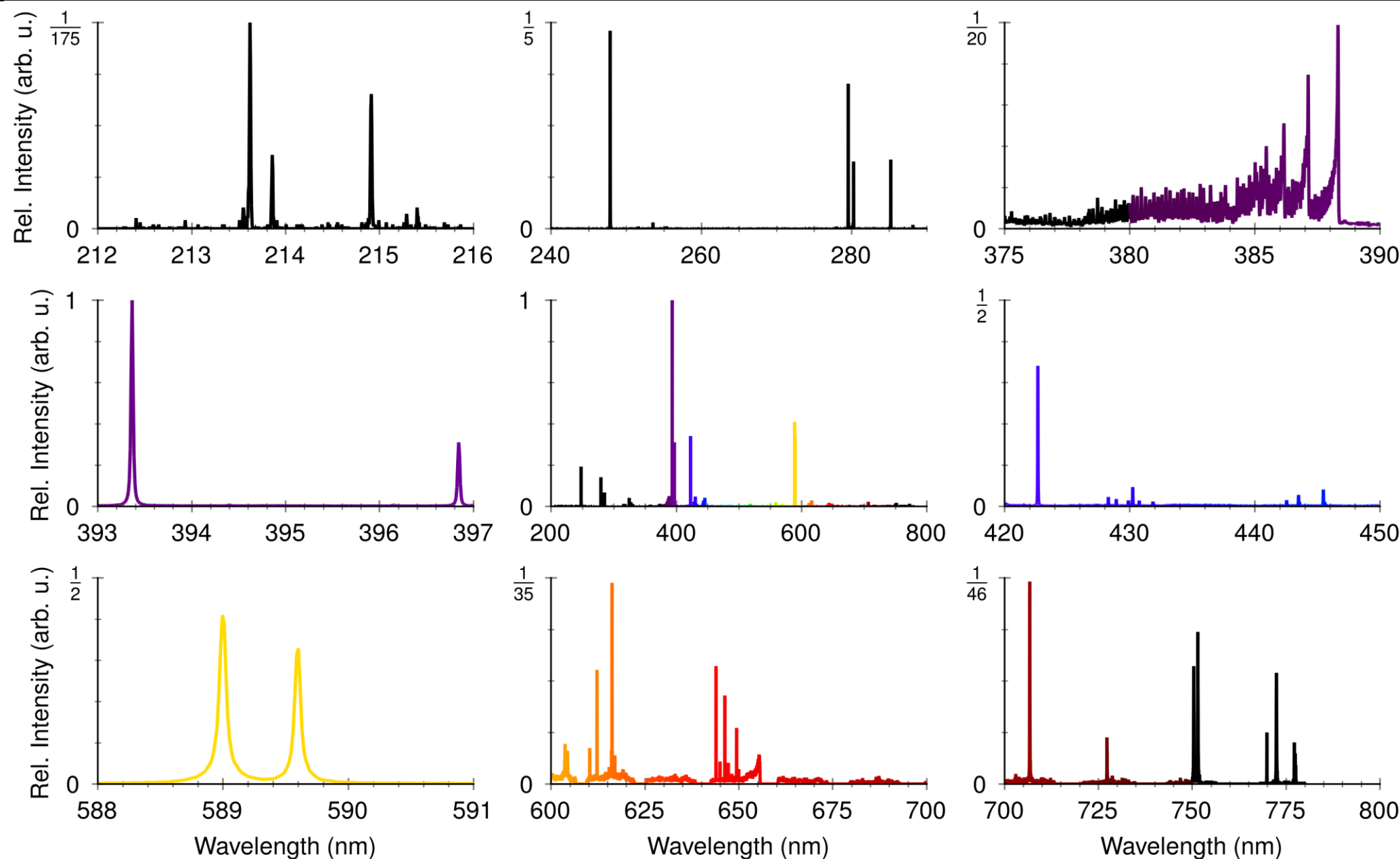
- Laser-induced breakdown threshold is higher in liquids
- Preference for UV laser
- Shockwave in liquid

Centrifugation cone helps concentrate the bacteria in each aCSF solution onto the nitrocellulose substrate, so **more cells are ablated per shot**.

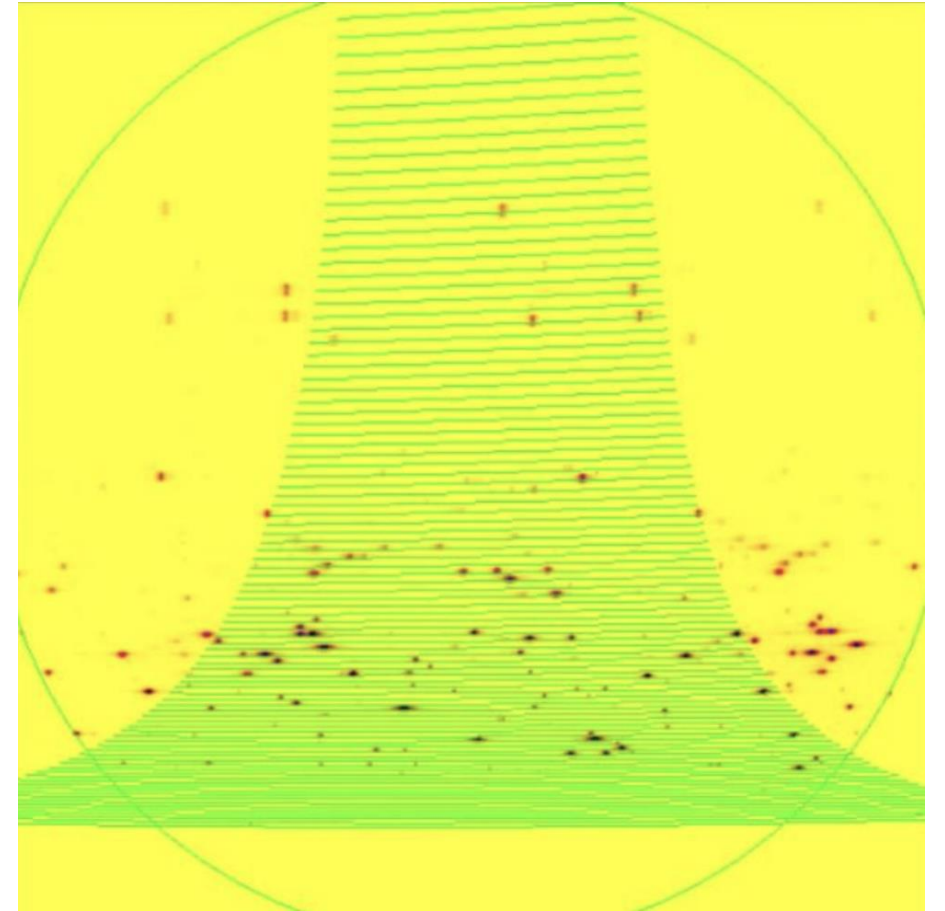
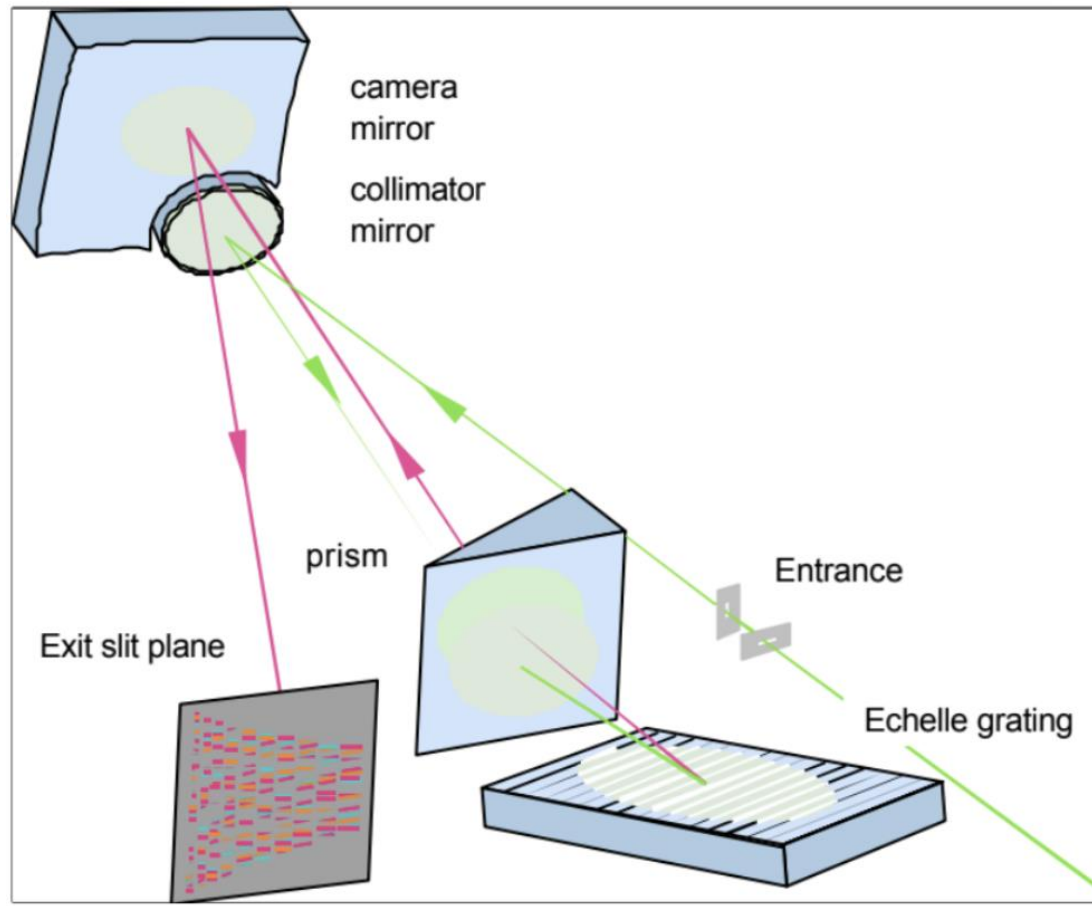
Broadening of Spectral Lines

1. Thermal Doppler broadening
2. Natural line broadening
3. Collision broadening → Stark broadening

Features in the LIBS Spectrum



Échelle Optics



Courtesy of LLA Instruments, GmbH